

# Aayush Bathija

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## EDUCATION

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### Oak Park High School

Oak Park, CA

Expected graduation 2029.

### Moorpark College, dual enrollment

2022–present

Calculus I–II, Trigonometry, College Algebra, General Physics I, Introduction to Symbolic Logic, Introduction to Philosophy, Introduction to Psychology.

**Independent study.** Real analysis through Tao's *Analysis I*.

## RESEARCH

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### Bounded ratios for Lorentzian polynomials

2026–present

Pushing the Huang–Huh–Soskin–Wang results into degree three. Manuscript in preparation with Prof. Daniel Soskin (UCLA) and one other coauthor. Funded by ORMC's Vertical Research Integration program.

### Fairness of voting systems

2025–present

Simulating ranked-choice, STAR, and related methods to compare how each one fails. Started in the Non-Trivial fellowship and still running.

## TEACHING

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### Instructor, UCLA Olga Radko Math Circle

2026–present

Teach a weekly section, and take the Olympiad course.

### Instructor, Agoura Math Circle

2024–2026

Taught and graded weekly classes. Primary on the documentation team, and wrote the Apps Script that migrates and reformats the curriculum archive each session.

### Volunteer coach, Medea Creek Middle School

2025–present

Run practice for the school's MATHCOUNTS team.

## LEADERSHIP

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- President, Philosophy Club, Oak Park High School. I pick the readings, write the presentations, and have moved the club toward logic and analytic philosophy
- Founder, pure mathematics club, Oak Park High School. Proof-based material for students who have run out of competition problems to be interested in

## AWARDS

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- Guinness World Record, most countries identified by outline in one minute
- John Locke Institute Essay Competition, shortlisted in 2025 and 2026
- Superintendent's Compassionate and Creative Citizens Award, given to four students district-wide and chosen by the superintendent
- Great History Challenge, national runner-up
- Berkeley Math Tournament, top ten in the Power Round (team)
- AMC 10A, 118.5 in 2026 (AIME qualifier)

## PROGRAMS

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### AwesomeMath Academy

2024–present

Number Theory with Mirroslav Yotov, into algebraic number theory: Gaussian integers, unique factorization, quadratic residues. Also Computational Geometry and Counting Strategies.

### Non-Trivial Research Fellowship

2025

Where the voting systems project started.

### LEAF Program, *The Mathematics of Morality*

2024

## OTHER

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I run *Academic Learning* (youtube.com/@Academic-Learning), where I work through whatever I am reading on camera. Speech and debate: reached the final round of the state qualifier. Speedcubing: 6.58s single,  $3 \times 3$ .